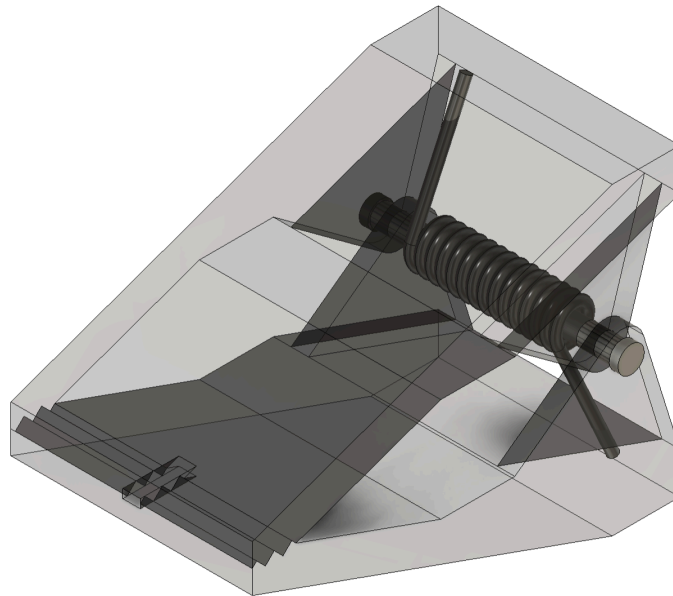


## **Project 4: Chip Clip**

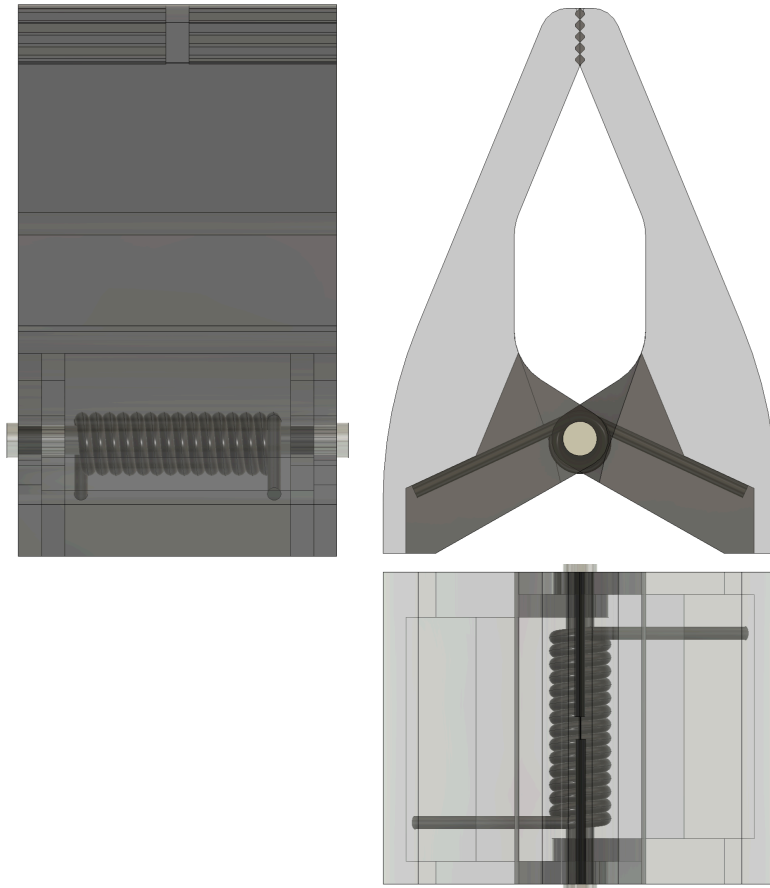
Machine Elements and Fatigue Design MAE 3620

April 27th, 2026

Anthony Bartolotti, Ty Davin, Matthew Herring, Patrick Jacobs



## Executive Summary



**Figure 1.** A top, front, and side view of our full clip assembly.

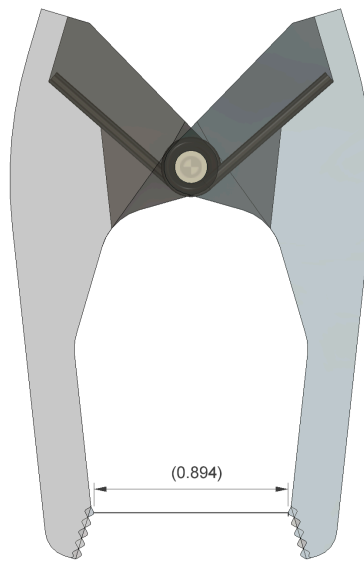
## Introduction

This project required designing a torsion spring and clip assembly for a chip clip. The spring must operate between 1.5 in-lbs preload and a 3.0 in-lbs maximum moment while achieving a minimum 5000-cycle life. Geometric constraints include a  $2.50" \times 1.75" \times 1.5"$  packaging envelope, 0.313" maximum spring outer diameter, 0.750" maximum lead length, and 1.000" maximum spring width. The clip must open beyond 1.000" when fully loaded. The material we selected for our spring is A228 music wire, the clip body is ABS plastic, and the pin is 304 stainless steel. Our design exhibits infinite life, a maximum opening of 1.295", and does not exceed 1" until it is within the 2.5 to 3 in-lbs range while satisfying all other constraints.

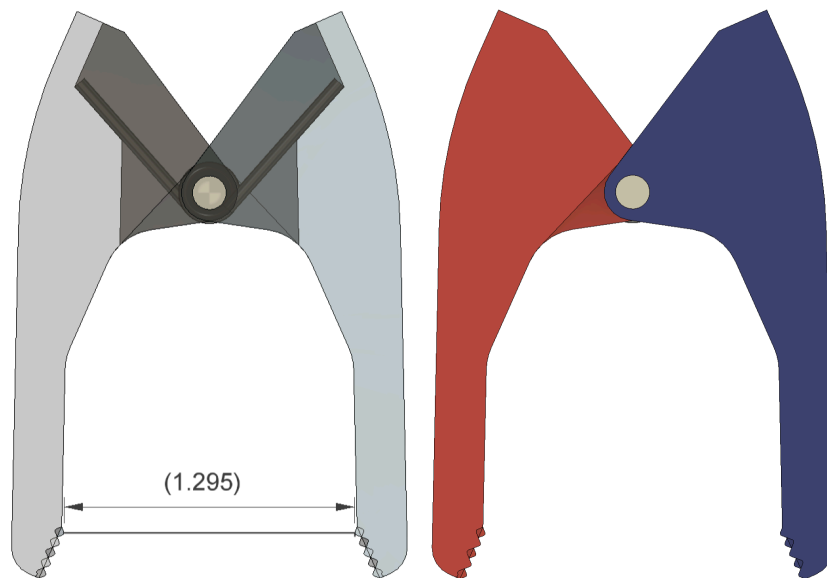
## Considerations

### Spring Constant

The moment will remain between 1.5 and 3.0 in-lbs as the pretension on the spring is 1.5 in-lbs, ensuring the moment does not fall below the range, and there is a physical stop on the spring that limits the displacement so that when the moment reaches 2.95 in-lbs the spring can no longer be displaced, ensuring the moment does not exceed the range.



**Figure 2.** The clip's opening distance with 2.5 in-lbs applied.



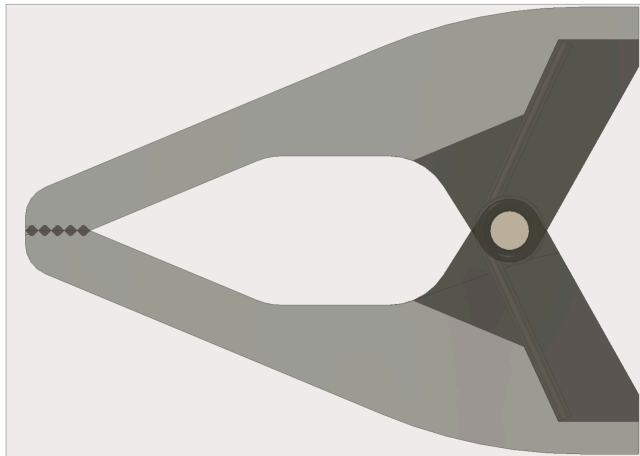
**Figure 3.** The clip's opening distance with 2.95 in-lbs applied where it hits a physical stop as seen above.

### **Loading**

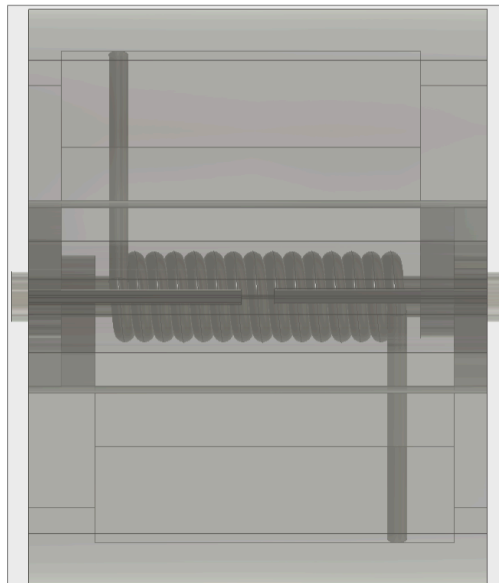
The spring is to be manufactured with 14 coils. When the spring is installed within the clip, it will be preloaded to 14.137 coils, and when fully loaded, it will have 14.27 coils. Associated calculations can be found in Appendix C below.

### **Geometry**

The clip has maximum dimensions of 2.375" x 1.73" x 1.475", which fits entirely in the 2.50" x 1.75" x 1.50" bounding box specification.



**Figure 4.** x-y Bounding box (2.50" x 1.75")



**Figure 5.** y-z Bounding box (1.75" x 1.50")

## **Stress**

The spring is designed with infinite life, and so it will not break under cyclic loading, nor will it yield under full opening. Associated calculations can be found in Appendix C below.

## **Pin**

The pin will not be pinched by the spring when the maximum moment is applied; the associated calculations can be found in Appendix C below. The deformation due to the spring pressure is negligible because we have a 1/8" 304 Stainless Steel pin. The pin can be thought of as a beam with a center load, which would have a maximum deflection of  $\delta = \frac{FL^3}{48EI}$ . The length is 0.975", E is 29 Mpsi, I is 0.0000479 in<sup>4</sup>, and the maximum force cannot be greater than 4 lbs, as the moment displacing the spring is 3 in-lbs and our lead length is 0.75". Given all of this, the deflection of the pin would be 0.0027 inches, which we deemed insignificant, as the difference in the spring's minimum diameter of 0.152" and the pin diameter is an order of magnitude greater.

## **Clip**

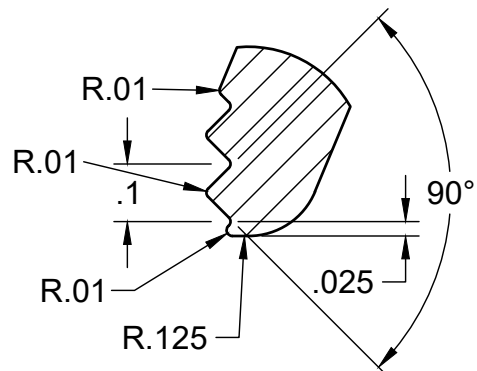
The clip has a maximum opening of 1.295" with an applied moment of 2.95 in-lbs. The clip cannot open further as a mechanical stop is in place, preventing additional angular deflection under any greater load. This stop was incorporated by rotating the clip to the angle that corresponds to 2.95 in-lbs and adding a section of material to the existing clip arms that will interfere with any rotation beyond this point. This stop can be seen above in Figure 3, more visible in the red and blue assembly where it is just above the pin.

## **Summary**

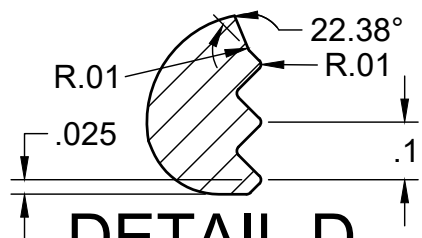
All design specifications were met. The spring achieves infinite fatigue life, exceeding the 5000 cycle minimum. The clip fits within the packaging envelope with a minimum of 0.02" of clearance. Opening the clip beyond 1" requires a moment exceeding 2.5 in-lbs, and the maximum moment that can be applied is 2.95 in-lbs, staying within the moment specifications.

To improve the spring further, several alternative design choices could be made. By increasing the wire diameter, the spring will fatigue more slowly, increasing the lifespan of the spring further. By expanding the moment range specifications we designed around, the spring could be made weaker which would ultimately allow us to use less material and make the clip require less force to use. Lastly, by using a standardized torsion spring from a supplier rather than a fully custom spring, the cost of production for this chip clip would be significantly less.

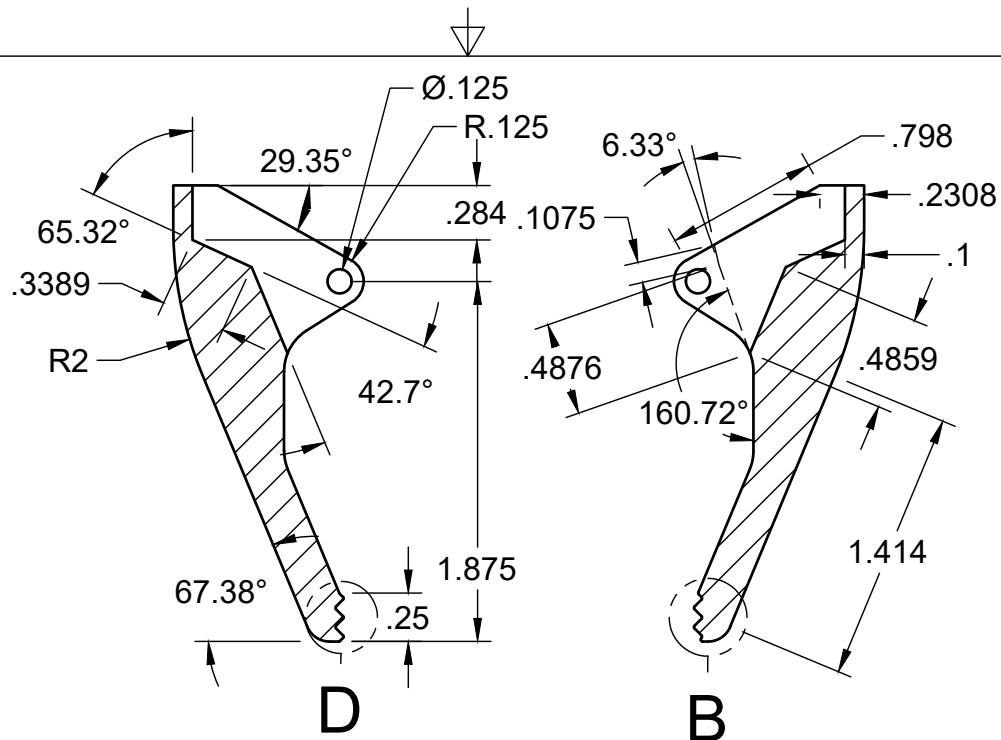
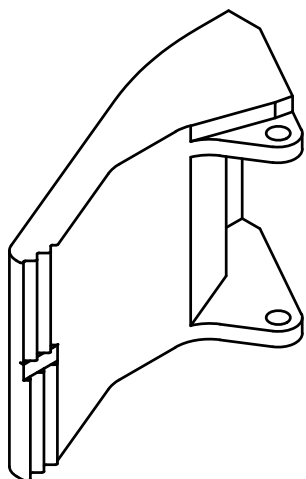
## **Appendix A**



**DETAIL B**  
**SCALE 3:1**  
Tolerance:  $\pm 0.001$

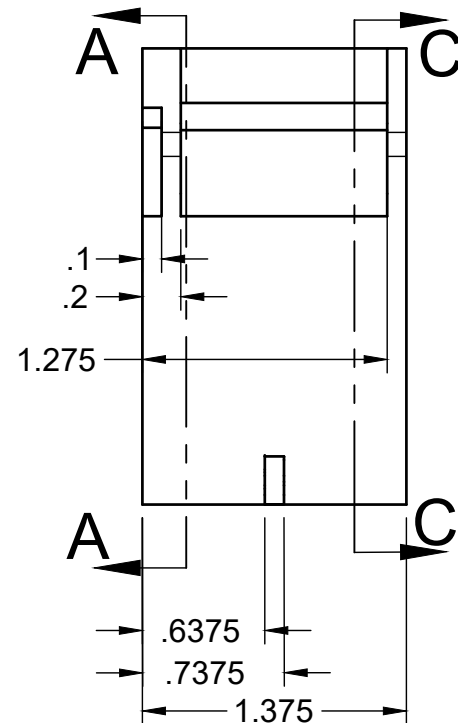


**DETAIL D**  
**SCALE 3:1**  
Tolerance:  $\pm 0.001$

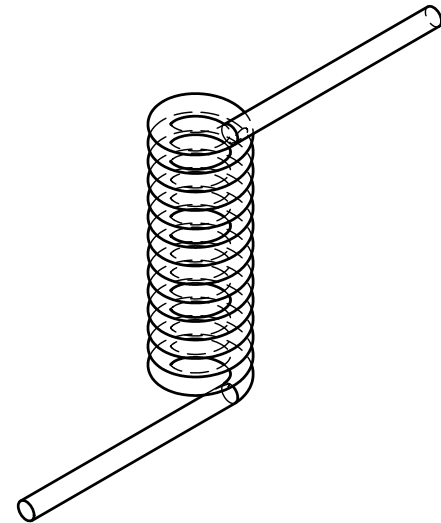
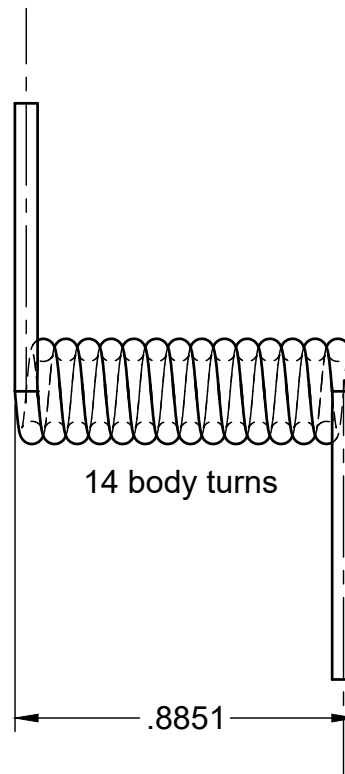
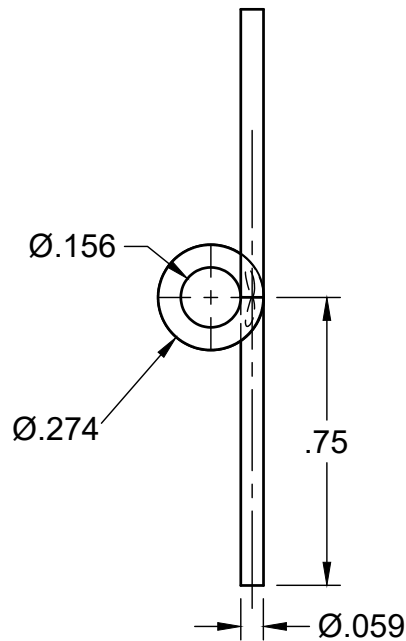


**SECTION C-C**  
**SCALE 1:1**

**SECTION A-A**  
**SCALE 1:1**



|                                                                            |  |   |                          |        |           |
|----------------------------------------------------------------------------|--|---|--------------------------|--------|-----------|
| Part Number                                                                |  | 1 |                          |        |           |
| UNLESS OTHERWISE SPECIFIED:<br>Units: Inches<br>Tolerance = $\pm 0.005$ in |  |   | PROJECT                  |        |           |
|                                                                            |  |   | Machine Elem - Chip Clip |        |           |
| APPROVED                                                                   |  |   | TITLE                    |        |           |
|                                                                            |  |   | Clip Arm                 |        |           |
| CHECKED                                                                    |  |   | SIZE                     | CODE   | DWG NO    |
| DRAWN                                                                      |  |   | A                        |        |           |
| Patrick Jacobs 4/27/2026                                                   |  |   | SCALE 1:1                | WEIGHT | SHEET 2/3 |



|                                                                           |                          |
|---------------------------------------------------------------------------|--------------------------|
| Part Number                                                               | 2                        |
| UNLESS OTHERWISE SPECIFIED:<br>Units: Inches<br>Tolerance = $\pm .001$ in |                          |
| Material                                                                  |                          |
| Music Wire                                                                |                          |
| APPROVED                                                                  |                          |
| CHECKED                                                                   |                          |
| DRAWN                                                                     | Patrick Jacobs 4/27/2026 |

PROJECT

**Machine Elem - Chip Clip**

TITLE

**Unloaded Spring**

SIZE

A

CODE

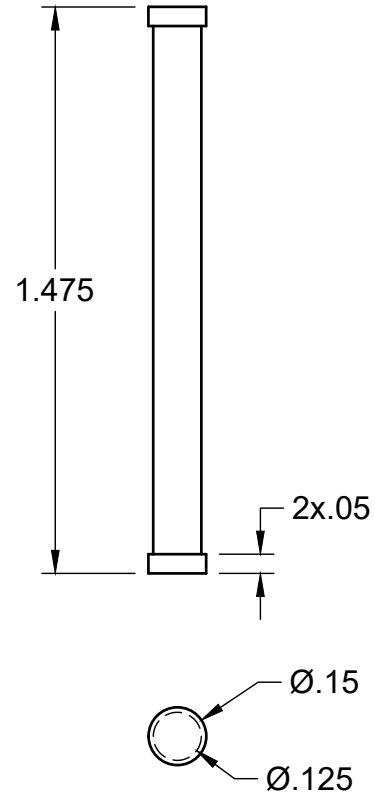
DWG NO

REV

SCALE 2:1

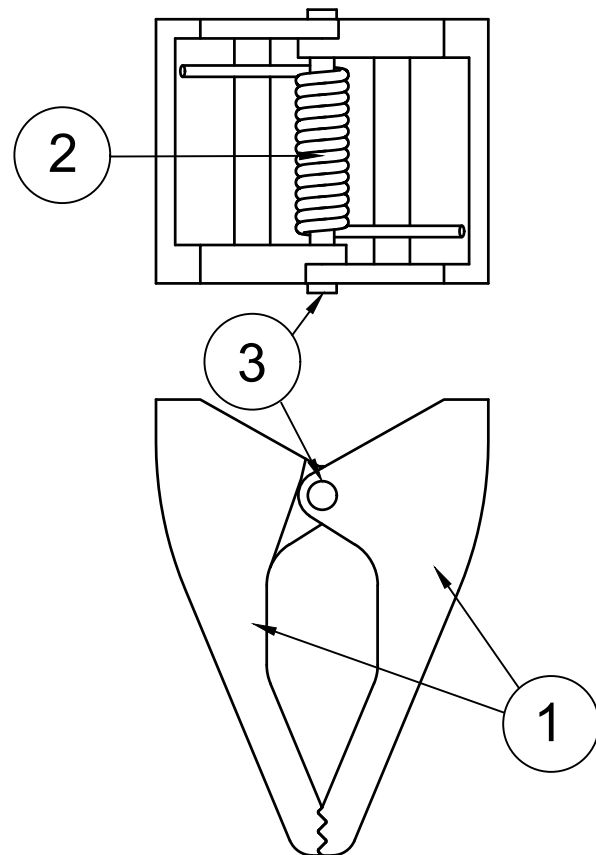
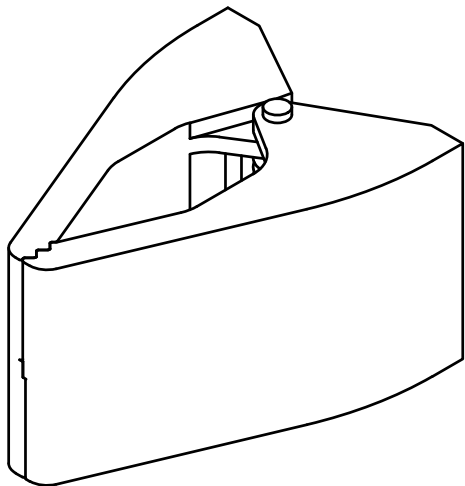
WEIGHT

SHEET 1/1



|                                                                      |  |                          |           |
|----------------------------------------------------------------------|--|--------------------------|-----------|
| Part Number                                                          |  | 3                        |           |
| UNLESS OTHERWISE SPECIFIED:<br>Units: Inches<br>Tolerance = ±.005 in |  | PROJECT                  |           |
|                                                                      |  | Machine Elem - Chip Clip |           |
|                                                                      |  | TITLE                    |           |
|                                                                      |  | Pin                      |           |
| Material                                                             |  |                          |           |
| 304 Stainless Steel                                                  |  |                          |           |
| APPROVED                                                             |  | SIZE                     | CODE      |
| CHECKED                                                              |  | A                        | DWG NO    |
| DRAWN                                                                |  | Patrick Jacobs           | 4/27/2026 |
| SCALE                                                                |  | 2:1 (From parent)        | WEIGHT    |
| SHEET                                                                |  | 3/3                      |           |

## **Appendix B**



#### Parts List

| Name     | Number |
|----------|--------|
| Clip Arm | 1      |
| Spring   | 2      |
| Pin      | 3      |

PROJECT

**Machine Elem - Chip Clip**

TITLE

**General Assembly**

APPROVED

CHECKED

DRAWN Patrick Jacobs 4/27/2026

SIZE

A

CODE

DWG NO

REV

SCALE 1:1

WEIGHT

SHEET 1/3

## Appendix C

Loading Calculations:

$$L_1 = L_2 = L_e = 0.750''$$

$$E = 29 \cdot 10^6 \text{ psi}$$

$$M_{\text{preload}} = 1.5 \text{ in/lbs}, M_{\text{minopen}} = 2.5 \text{ in/lbs}, M_{\text{maxopen}} = 2.95$$

$$25 \text{ gauge music wire + tolerance } d = 0.059 + 0.001 = 0.060''$$

$$D = 0.215 + 0.001 = 0.216''$$

$$D_o = D + d = 0.276''$$

$$N_b = 14$$

$$N_a = N_b + \frac{2 \cdot L_e}{3\pi D} = 14 + \frac{2 \cdot 0.75}{3\pi(0.216)} = 14.7368$$

$$C = \frac{D}{d} = \frac{0.216}{0.060} = 3.6$$

$$\theta = \frac{67.8 \cdot M \cdot D}{d^4 \cdot E} \left( N_b + \frac{2 \cdot L_e}{3\pi D} \right) \cdot \frac{180}{\pi}$$

$$\theta_{\text{preload}} = \frac{67.8(1.5)(0.216)}{(0.060)^4(29 \cdot 10^6)} \left( 14 + \frac{2(0.75)}{3\pi(0.216)} \right) \cdot \frac{180}{\pi} = 49.351^\circ$$

$$\text{i.e. } 130.649^\circ$$

$$\phi_{49.351^\circ}$$

$$= 0.137$$

$$N'_b = 14.137 \text{ coils}$$

$$\theta_{\text{minopen}} = \frac{67.8(2.5)(0.216)}{(0.060)^4(29 \cdot 10^6)} \left( 14 + \frac{2(0.75)}{3\pi(0.216)} \right) \cdot \frac{180}{\pi} = 82.252^\circ$$

$$\text{i.e. } 97.748^\circ$$

$$\phi_{82.252^\circ}$$

$$\theta_{\text{maxopen}} = \frac{67.8(2.95)(0.216)}{(0.060)^4(29 \cdot 10^6)} \left( 14 + \frac{2(0.75)}{3\pi(0.216)} \right) \cdot \frac{180}{\pi} = 97.058^\circ$$

$$= 0.270$$

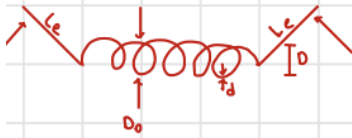
$$\text{i.e. } 82.742^\circ$$

$$\phi_{97.058^\circ}$$

$$N''_b = 14.27 \text{ coils}$$

## Stress Calculations:

Fatigue factor of safety Spring:



$$L_1 = L_2 = L_e \rightarrow L_e = 0.75''$$

$$E = 29 \times 10^6 \text{ psi}$$

$$M_{\text{preload}} = 1.5 \text{ in lb}$$

$$\text{AS gauge music wire} \rightarrow d = 0.039'' \rightarrow \text{w/ worsening tolerance: } d = 0.039'' - 0.001''$$

$$d = 0.038''$$

$$D = 0.213'' \rightarrow \text{w/ worsening tolerance: } D = 0.213'' - 0.001''$$

$$D = 0.214''$$

$$D_o = D + d = 0.214'' + 0.038''$$

$$D_o = 0.272''$$

$$\text{Body turns} \rightarrow N_b = 14$$

$$\text{Ultimate stress: } S_{ut} = \frac{A}{d^m}$$

Table 10-4:

$$A = 801 \times 10^3 \text{ psi in}^m$$

$$m = 0.145$$

$$S_{ut} = \frac{801 \times 10^3 \text{ psi in}^m}{(0.038'')^{0.145}}$$

$$S_{ut} = 303737.90866 \text{ psi}$$

$$S_r = 0.5 S_{ut} = 0.5 (303737.90866 \text{ psi})$$

$$S_r = 151868.95433 \text{ psi}$$

$$\text{Yield strength: } S_y = 0.78 S_{ut}$$

$$S_y = 0.78 (303737.90866 \text{ psi})$$

$$S_y = 236915.56875 \text{ psi}$$

Spring index C:

$$C = \frac{D}{d} = \frac{0.214''}{0.038''} = 3.68965519241$$

$$\text{Stress correction factor: } k_i = \frac{4C^2 - C - 1}{4C(C-1)}$$

$$k_i = \frac{4C^2 - C - 1}{4C(C-1)} = \frac{4(3.68965519241)^2 - 3.68965519241 - 1}{4(3.68965519241)(3.68965519241 - 1)} = 1.23863444524$$

Gerber criterion for bending endurance strength:

$$S_e = \frac{0.5 S_r}{1 - \left(\frac{0.5 S_r}{S_{ut}}\right)^2}$$

$$S_e = \frac{0.5 (151868.95433 \text{ psi})}{1 - \left(\frac{0.5 (151868.95433 \text{ psi})}{303737.90866 \text{ psi}}\right)^2}$$

$$S_e = 80996.7786427 \text{ psi}$$

Closed Stress:

$$\sigma_{\text{closed}} = k_i \cdot \frac{32 M_{\text{preload}}}{\pi d^3}$$

$$\sigma_{\text{closed}} = 1.8265444524 \cdot \frac{32(1.5 \text{ in lb})}{\pi(0.0531)^3}$$

$$\sigma_{\text{closed}} = 98171.4552739 \text{ psi}$$

Opening Stress:  $\rightarrow \text{Max moment} = \text{opening moment} = 3 \text{ in lb}$

$$\sigma_{\text{open}} = k_i \cdot \frac{32 M_{\text{max}}}{\pi d^3}$$

$$\sigma_{\text{open}} = 1.8265444524 \cdot \frac{32(3 \text{ in lb})}{\pi(0.0531)^3}$$

$\rightarrow \text{Note use worst case scenario } 3 \text{ in lb even though we have stop before } 3 \text{ in lb}$

$$\sigma_{\text{open}} = 196342.910548 \text{ psi}$$

Mean Stress:

$$\sigma_m = \frac{\sigma_{\text{open}} + \sigma_{\text{closed}}}{2} = \frac{(98171.4552739 \text{ psi}) + (196342.910548 \text{ psi})}{2}$$

$$\sigma_m = 147207.182911 \text{ psi}$$

Alternating Stress:

$$\sigma_a = \sigma_m - \sigma_{\text{closed}}$$

$$\sigma_a = 147207.182911 \text{ psi} - 98171.4552739 \text{ psi}$$

$$\sigma_a = 49035.7276369 \text{ psi}$$

Gerber yielding factor of safety:

$$n_f = 0.5 \left( \frac{\sigma_a}{S_e} \right) \left( \frac{S_{ut}}{\sigma_m} \right)^2 \left[ -1 + \sqrt{1 + \left( \frac{2 \sigma_m S_e}{S_{ut} \sigma_a} \right)^2} \right]$$

$$n_f = 0.5 \left( \frac{49035.7276369 \text{ psi}}{80996.7756427 \text{ psi}} \right) \left( \frac{303737.90866 \text{ psi}}{147207.182911 \text{ psi}} \right)^2 \left[ -1 + \sqrt{1 + \left( \frac{2(147207.182911 \text{ psi})(80996.7756427 \text{ psi})}{(303737.90866 \text{ psi})(49035.7276369 \text{ psi})} \right)^2} \right]$$

$$n_f = 1.14221091972 \rightarrow \text{infinite life } n_f > 1 \rightarrow \text{works even w/ worst tolerance}$$

Yielding factor of safety using larger lines:

$$\frac{1}{n_y} = \frac{\sigma_a + |\sigma_m|}{S_y}$$

$$n_y = \left( \frac{\sigma_a + |\sigma_m|}{S_y} \right)^{-1}$$

$$n_y = \left( \frac{49035.7276369 \text{ psi} + 147207.182911 \text{ psi}}{226915.568755 \text{ psi}} \right)^{-1}$$

$$n_y = 1.20664182931 < 1 \rightarrow \text{will not yield}$$

Pin Calculations:

$$\theta'_c = \frac{10.8 \cdot M_{\max open} \cdot D \cdot N_b}{d^4 E} = \frac{10.8(2.995)(0.216)(14)}{(0.060)^4(29 \cdot 10^6)} = 0.260 \text{ revs}$$
$$D_p = \frac{1}{8}''$$
$$\Delta = \frac{N_b D}{N_b + \theta'_c} - d - D_p = \frac{14(0.216)}{14 + 0.260} - 0.060 - \frac{1}{8}$$
$$\Delta = 0.027'' > 0.001''$$

# Preliminary Calculations

## Part I

### Diameter Selection

Graphing to select a d and D

#### 1 Governing Equations

**Note:** The following equations are expressed solely in terms of d and D, and can be used to create restrictions on the values for d and D.

**Gerber Equation:**

$$n_f = 0.5 \left( \frac{\sigma_a}{S_e} \right) \left( \frac{S_{ut}}{\sigma_m} \right)^2 \left( -1 + \sqrt{1 + \left( \frac{2\sigma_m S_e}{S_{ut}\sigma_a} \right)^2} \right)$$

**Spring Length:**

$$L_s = N_b \cdot d$$

**Fit on Shaft:**

$$\Delta = \frac{N_b D}{N_b + \theta_c} - d - D_p$$

**Yielding Safety Factor:**

$$n_y = \left( \frac{\sigma_a + \sigma_m}{\sigma_{open}} \right)^{-1}$$

#### 2 Graphing

Graph a region using the governing equations such that:

$$D = x$$

$$d = y$$

$$\{1 < n_f < 1.5\} \{L_s < 1\} \{D > 0\} \{d > 0\} \{D + d < 0.313\} \{\Delta > 0.02\} \{n_y > 1.2\}$$

To help visualize this section graph a target number of body coils as well as a target wire gauge:

$$N_{btarget} = 14$$

$$y = \left( \left( N_{btarget} + \frac{L_1 + L_2}{67.8 \cdot D \cdot N_a} \right) \left( \frac{E}{67.8D} \cdot \frac{\theta_{open}}{M} \right)^{-1} \right)^{\frac{1}{4}}$$

$$G_d = 0.059'' \text{ (25 gauge)}$$

$$y = G_d = 0.059$$

The grey region is defined by the inequalities above, the red line shows the target body coils,  $N_{btarget}$ , and the purple line shows the target gauge,  $G_d$ . By selecting a point in this region a spring can be made that should meet all criteria, which can be individually checked later. The lines are to make selecting a point easier as they give a reference.

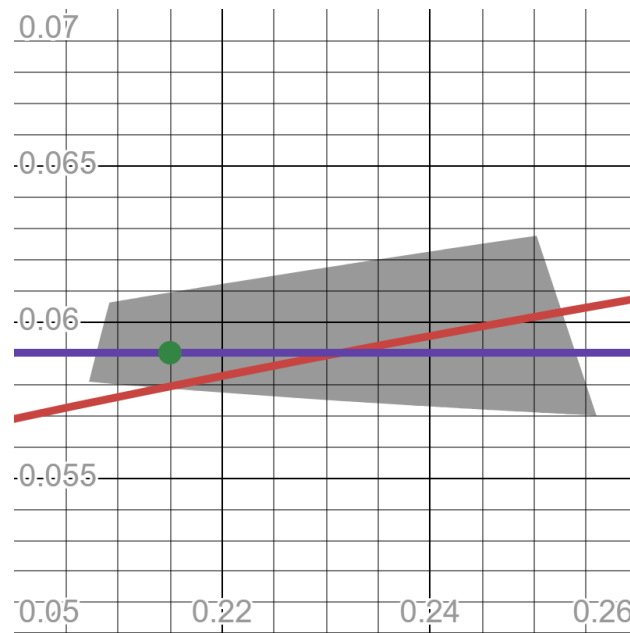


Figure C-1: Desmos Drawing

### 3 Point

From Figure C-1 the green point was chosen, which is described by the following:

$$D_1 = 0.215''$$

$$d_1 = 0.059''$$

Which results in the following approximate numbers:

$$N_{a1} = \frac{d_1^4 \cdot 29 \cdot 10^6}{67.8 D_1} \cdot \frac{\theta_{open}}{M} = 11.39 \text{ turns}$$

$$N_{b1} = N_{a1} - \frac{L_1 + L_2}{67.8 \cdot D_1 \cdot N_{a1}} = 11.38 \text{ turns}$$

$$L_{s1} = N_{b1} \cdot d_1 = 0.6258''$$

$$C_1 = \frac{D_1}{d_1} = 3.91$$

These numbers can be used to create the actual spring values, which if they are selected close to these the spring should pass all requirements.

## Part II

# Supporting Equations

Equations that relate each Governing Equation to d and D

## 4 Geometry

**Fulcrum Length:**

$$L_f = 0.625''$$

**Arm Length:**

$$L_a = 1.5''$$

**Hypotenuse Length:**

$$L_h = \sqrt{L_f^2 + L_a^2} = \sqrt{0.625''^2 + 1.5''^2} = 1.625''$$

**Open Length:**

$$L_o = 1.0''$$

**Angle  $\gamma$ :**

$$\gamma = \arcsin\left(\frac{\frac{L_o}{2}}{L_h}\right) = \arcsin\left(\frac{\frac{1''}{2}}{1.625''}\right) = 0.3128 \text{ rad}$$

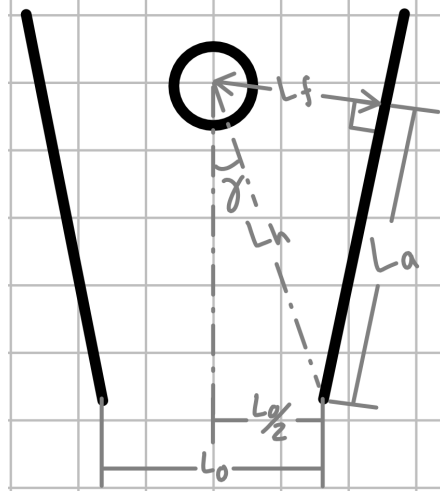


Figure C-2: Geometry Drawing

**Lead Lengths:**

$$L_1 = L_2 = 0.75''$$

**Pin Diameter:**

$$D_p = \frac{1}{8}''$$

## 5 Material Properties (In Terms of d)

**Note:** Music Wire (A228), not shot peened

**Modulus of Elasticity:** (Table 10-5)

$$E = \begin{cases} 29.5 \cdot 10^6 \text{ psi} & d < 0.032'' \\ 29.0 \cdot 10^6 \text{ psi} & 0.033'' < d < 0.063'' \\ 28.5 \cdot 10^6 \text{ psi} & 0.064'' < d < 0.15'' \\ 28.0 \cdot 10^6 \text{ psi} & 0.125'' < d \end{cases}$$

**Ultimate Strength:** (Table 10-4)

$$A = 201 \cdot 10^3 \text{ psi}$$

$$m = 0.145$$

$$S_{ut} = \frac{A}{d^m}$$

**Recommended Bending Stress:** (Table 10-10)

$$S_r = 0.5 \cdot S_{ut}$$

**Endurance Strength:**

$$S_e = \frac{0.5S_r}{1 - \left(\frac{0.5S_r}{S_{ut}}\right)^2}$$

**Static Strength:**

$$S_y = 0.78 \cdot S_{ut}$$

## 6 Stress Supporting Calculations

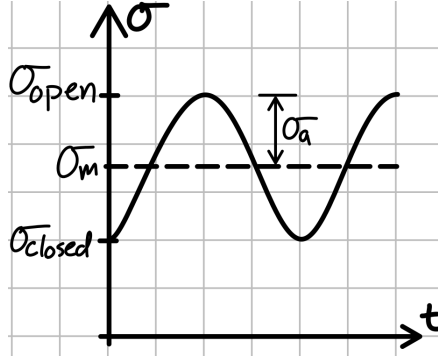


Figure C-3: Stress Drawing

**Angle When Open:**

$$\theta_{open} = 2\gamma = 0.6255 \text{ rad}$$

**Total Load:**

$$M_{total,min} = 2.5 \text{ in} \cdot \text{lb}$$

$$M_{total,max} = 3.0 \text{ in} \cdot \text{lb}$$

**Preload:**

$$M_{pre} = 1.5 \text{ in} \cdot \text{lb}$$

**Applied Load:**

$$M_{open} = M_{total,min} - M_{pre} = 2.5 \text{ in} \cdot \text{lb} - 1.5 \text{ in} \cdot \text{lb} = 1.0 \text{ in} \cdot \text{lb}$$

$$M_{max} = M_{total,max} - M_{pre} = 3.0 \text{ in} \cdot \text{lb} - 1.5 \text{ in} \cdot \text{lb} = 1.5 \text{ in} \cdot \text{lb}$$

**Total Turns:**

$$N_a = \frac{d^4 E}{67.8 D} \cdot \frac{\theta_{open}}{M_{open}}$$

**Body Turns:**

$$N_b = N_a - \frac{L_1 + L_2}{3\pi \cdot D}$$

**Spring Index:**

$$C = \frac{D}{d}$$

**Interior Stress Correction Factor:**

$$K_i = \frac{4C^2 - C - 1}{4C(C - 1)}$$

**Stress When Closed:**

$$\sigma_{closed} = K_i \cdot \frac{32M_{pre}}{\pi d^3}$$

**Stress When Open:**

$$\sigma_{open} = K_i \cdot \frac{32(M_{pre} + M_{max})}{\pi d^3}$$

**Mean Stress:**

$$\sigma_m = \frac{\sigma_{closed} + \sigma_{open}}{2}$$

**Alternating Stress:**

$$\sigma_a = \sigma_{open} - \sigma_m$$

## 7 Geometry Supporting Calculations

**Coil Deflection:**

$$\theta_c = \frac{10.8 \cdot (M_{max} + M_{pre}) \cdot D \cdot N_b}{d^4 \cdot E}$$